

## 31. EUSART - Enhanced Universal Synchronous Asynchronous Receiver Transmitter

The Enhanced Universal Synchronous Asynchronous Receiver Transmitter (EUSART) module is a serial I/O communications peripheral. It contains all the clock generators, shift registers and data buffers necessary to perform an input or output serial data transfer independent of device program execution. The EUSART, also known as a Serial Communications Interface (SCI), can be configured as a full-duplex asynchronous system or half-duplex synchronous system.

**Table 31-1.** EUSART Module Availability

| Device      | EUSART1 |
|-------------|---------|
| PIC16F13113 | Yes     |
| PIC16F13114 | Yes     |
| PIC16F13115 | Yes     |
| PIC16F13123 | Yes     |
| PIC16F13124 | Yes     |
| PIC16F13125 | Yes     |
| PIC16F13143 | Yes     |
| PIC16F13144 | Yes     |
| PIC16F13145 | Yes     |

Full Duplex mode is useful for communications with peripheral systems, such as CRT terminals and personal computers. Half Duplex Synchronous mode is intended for communications with peripheral devices, such as A/D or D/A integrated circuits, serial EEPROMs or other microcontrollers. These devices typically do not have internal clocks for baud rate generation and require the external clock signal provided by a host synchronous device.

The EUSART module includes the following capabilities:

- Full-duplex asynchronous transmit and receive
- Two-character input buffer
- One-character output buffer
- Programmable 8-bit or 9-bit character length
- Address detection in 9-bit mode
- Input buffer overrun error detection
- Received character framing error detection
- Half-duplex synchronous host
- Half-duplex synchronous client
- Programmable clock polarity in Synchronous modes
- Sleep operation

The EUSART module implements the following additional features, making it ideally suited for use in Local Interconnect Network (LIN) bus systems:

- Automatic detection and calibration of the baud rate
- Wake-up on Break reception
- 13-bit Break character transmit

Block diagrams of the EUSART transmitter and receiver are shown in [Figure 31-1](#) and [Figure 31-2](#).

The operation of the EUSART module consists of six registers:

- The RXx/DTx and TXx/CKx input pins are selected with the RXxPPS and TXxPPS registers, respectively. TXx, CKx, and DTx output pins are selected with each pin's RxyPPS register. Since the RX input is coupled with the DT output in Synchronous mode, it is the user's responsibility to select the same pin for both of these functions when operating in Synchronous mode. The EUSART control logic will control the data direction drivers automatically.

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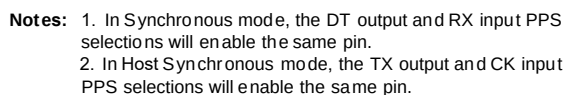
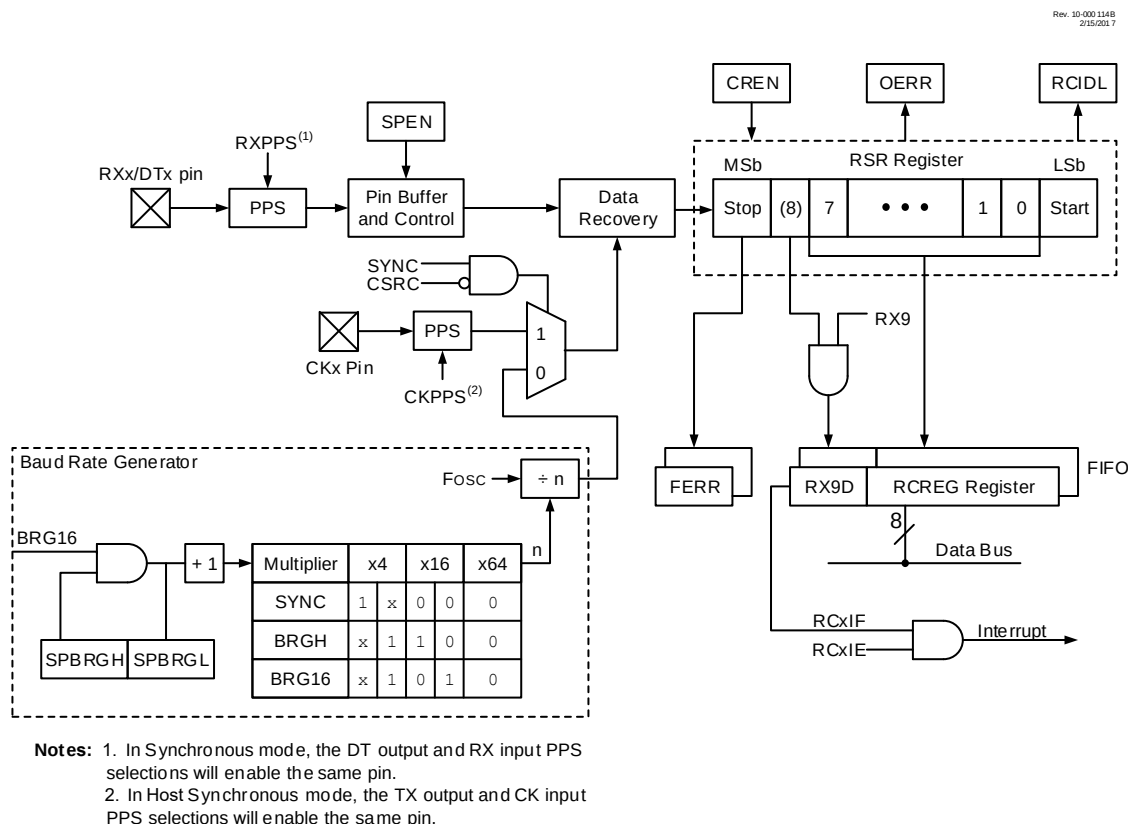


Figure 31-2. EUSART Receive Block Diagram



## 31.1. EUSART Asynchronous Mode

The EUSART transmits and receives data using the standard non-return-to-zero (NRZ) format. NRZ is implemented with two levels: a  $V_{OH}$  Mark state, which represents a '1' data bit, and a  $V_{OL}$  Space state, which represents a '0' data bit. NRZ refers to the fact that consecutively transmitted data bits of the same value stay at the output level of that bit without returning to a neutral level between each bit transmission. An NRZ transmission port idles in the Mark state. Each character transmission consists of one Start bit followed by eight or nine data bits and is always terminated by one or more Stop bits. The Start bit is always a space and the Stop bits are always marks. The most common data format is eight bits. Each transmitted bit persists for a period of  $1/(\text{Baud Rate})$ . An on-chip dedicated 8-bit/16-bit Baud Rate Generator is used to derive standard baud rate frequencies from the system oscillator. See [Table 31-3](#) for examples of baud rate configurations.

The EUSART transmits and receives the LSb first. The EUSART's transmitter and receiver are functionally independent, but share the same data format and baud rate. Parity is not supported by the hardware, but can be implemented in software and stored as the ninth data bit.

### 31.1.1. EUSART Asynchronous Transmitter

[Figure 31-1](#) is a simplified representation of the transmitter. The heart of the transmitter is the serial Transmit Shift Register (TSR), which is not directly accessible by software. The TSR obtains its data from the transmit buffer, which is the TXxREG register.

#### 31.1.1.1. Enabling the Transmitter

The EUSART transmitter is enabled for asynchronous operations by configuring the following three control bits:

- The Transmit Enable (TXEN) bit is set to '1' to enable the transmitter circuitry of the EUSART

- The EUSART Mode Select (**SYNC**) bit is set to '0' to configure the EUSART for asynchronous operation
- The Serial Port Enable (**SPEN**) bit is set to '1' to enable the EUSART interface and to enable automatically the output drivers for the RxyPPS selected as the TXx/CKx output

All other EUSART control bits are assumed to be in their default state.

If the TXx/CKx pin is shared with an analog peripheral, the analog I/O function must be disabled by clearing the corresponding ANSEL bit.



**Important:** The TXxIF Transmitter Interrupt Flag in the PIRx register is set when the TXEN enable bit is set and the Transmit Shift Register (TSR) is Idle.

### 31.1.1.2. Transmitting Data

A transmission is initiated by writing a character to the **TXxREG** register. If this is the first character, or the previous character has been completely flushed from the TSR, the data in the TXxREG is immediately transferred to the TSR register. If the TSR still contains all or part of a previous character, the new character data are held in the TXxREG until the Stop bit of the previous character has been transmitted. The pending character in the TXxREG is then transferred to the TSR in one  $T_{CY}$  immediately following the Stop bit transmission. The transmission of the Start bit, data bits and Stop bit sequence commences immediately following the transfer of the data to the TSR from the TXxREG.

### 31.1.1.3. Transmit Data Polarity

The polarity of the transmit data can be controlled with the Clock/Transmit Polarity Select (**SCKP**) bit. The default state of this bit is '0', which selects high true transmit Idle and data bits. Setting the SCKP bit to '1' will invert the transmit data resulting in low true Idle and data bits. The SCKP bit controls transmit data polarity in Asynchronous mode only. In Synchronous mode, the SCKP bit has a different function. See [Clock Polarity](#) for more details.

### 31.1.1.4. Transmit Interrupt Flag

The EUSART Transmit Interrupt Flag (TXxIF) bit of the PIRx register is set whenever the EUSART transmitter is enabled and no character is being held for transmission in the **TXxREG**. In other words, the TXxIF bit is only cleared when the TSR is busy with a character and a new character has been queued for transmission in the TXxREG. The TXxIF flag bit is not cleared immediately upon writing TXxREG. TXxIF becomes valid in the second instruction cycle following the write execution. Polling TXxIF immediately following the TXxREG write will return invalid results. The TXxIF bit is read-only, it cannot be set or cleared by software.

The TXxIF interrupt can be enabled by setting the EUSART Transmit Interrupt Enable (TXxIE) bit of the PIRx register. However, the TXxIF flag bit will be set whenever the TXxREG is empty, regardless of the state of TXxIE enable bit.

To use interrupts when transmitting data, set the TXxIE bit only when there is more data to send. Clear the TXxIE interrupt enable bit upon writing the last character of the transmission to the TXxREG.

### 31.1.1.5. TSR Status

The Transmit Shift Register Status (**TRMT**) bit indicates the status of the TSR register. This is a read-only bit. The TRMT bit is set when the TSR register is empty and is cleared when a character is transferred to the TSR register from the **TXxREG**. The TRMT bit remains clear until all bits have been shifted out of the TSR register. No interrupt logic is tied to this bit, so the user needs to poll this bit to determine the TSR status.



**Important:** The TSR register is not mapped in data memory, so it is not available to the user.

### 31.1.1.6. Transmitting 9-Bit Characters

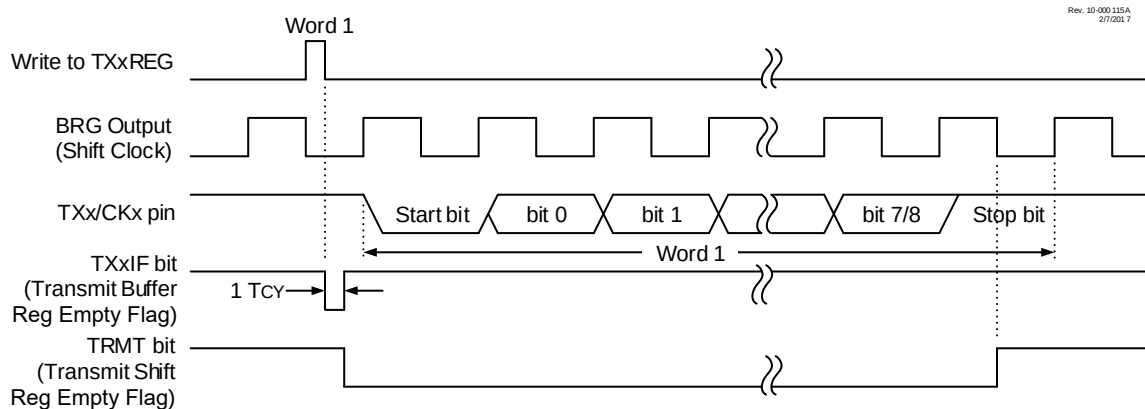
The EUSART supports 9-bit character transmissions. When the 9-Bit Transmit Enable (TX9) bit is set, the EUSART will shift nine bits out for each character transmitted. The TX9D bit is the ninth and Most Significant data bit. When transmitting 9-bit data, the TX9D data bit must be written before writing the eight Least Significant bits into the TXxREG. All nine bits of data will be transferred to the TSR register immediately after the TXxREG is written.

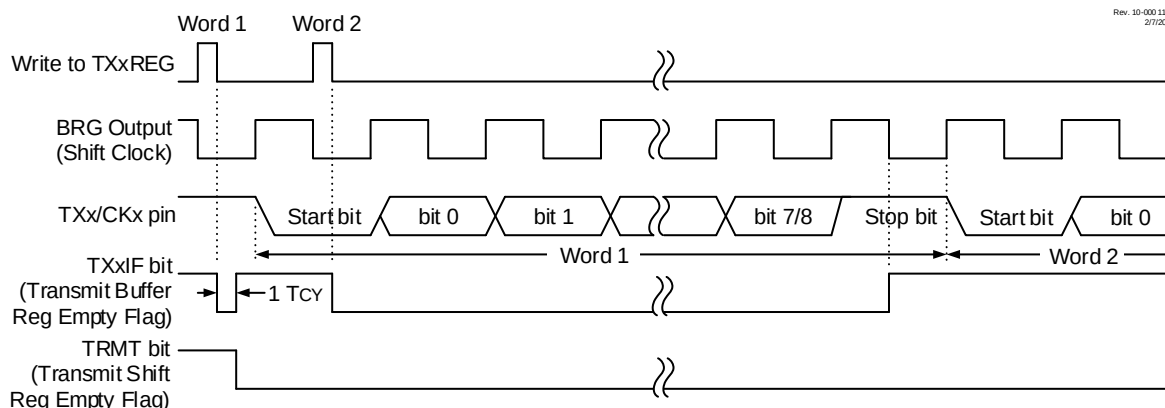
A special 9-bit Address mode is available for use with multiple receivers. See [Address Detection](#) for more information on the Address mode.

### 31.1.1.7. Asynchronous Transmission Setup

1. Initialize the SPxBRGH:SPxBRGL register pair and the BRGH and BRG16 bits to achieve the desired baud rate (see [EUSART Baud Rate Generator \(BRG\)](#)).
2. Select the transmit output pin by writing the appropriate value to the RxyPPS register.
3. Enable the asynchronous serial port by clearing the SYNC bit and setting the SPEN bit.
4. If 9-bit transmission is desired, set the TX9 control bit. That will indicate that the eight Least Significant data bits are an address when the receiver is set for address detection.
5. Set SCKP bit if inverted transmit is desired.
6. Enable the transmission by setting the TXEN control bit. This will cause the TXxIF interrupt bit to be set.
7. If interrupts are desired, set the TXxIE interrupt enable bit of the PIRx register.
8. An interrupt will occur immediately provided that the GIE and PEIE bits of the INTCON register are also set.
9. If 9-bit transmission is selected, the ninth bit will be loaded into the TX9D data bit.
10. Load 8-bit data into the TXxREG register. This will start the transmission.

**Figure 31-3.** Asynchronous Transmission



**Figure 31-4. Asynchronous Transmission (Back-to-Back)**

### 31.1.2. EUSART Asynchronous Receiver

The Asynchronous mode is typically used in RS-232 systems. A simplified representation of the receiver is shown in Figure 31-2. The data are received on the RXx/DTx pin and drive the data recovery block. The data recovery block is actually a high-speed shifter operating at 16 times the baud rate, whereas the serial Receive Shift Register (RSR) operates at the bit rate. When all eight or nine bits of the character have been shifted in, they are immediately transferred to a two character First-In-First-Out (FIFO) memory. The FIFO buffering allows reception of two complete characters and the start of a third character before software must start servicing the EUSART receiver. The FIFO and RSR registers are not directly accessible by software. Access to the received data is via the RCxREG register.

#### 31.1.2.1. Enabling the Receiver

The EUSART receiver is enabled for asynchronous operation by configuring the following three control bits:

- The Continuous Receive Enable (CREN) bit is set to '1' to enables the receiver circuitry of the EUSART
- The EUSART Mode Select (SYNC) bit is set to '0' to configure the EUSART for asynchronous operation
- The Serial Port Enable (SPEN) bit is set to '1' to enable the EUSART interface

All other EUSART control bits are assumed to be in their default state.

The user must set the RXxPPS register to select the RXx/DTx I/O pin and set the corresponding TRIS bit to configure the pin as an input.



**Important:** If the RX/DT function is on an analog pin, the corresponding ANSEL bit must be cleared for the receiver to function.

#### 31.1.2.2. Receiving Data

The receiver data recovery circuit initiates character reception on the falling edge of the first bit. The first bit, also known as the Start bit, is always a zero. The data recovery circuit counts one-half bit time to the center of the Start bit and verifies that the bit is still a zero. If it is not a zero, then the data recovery circuit aborts character reception without generating an error and resumes looking for the falling edge of the Start bit. If the Start bit zero verification succeeds, then the data recovery circuit counts a full bit time to the center of the next bit. The bit is then sampled by a majority detect circuit and the resulting '0' or '1' is shifted into the RSR. This repeats until all data bits have been sampled and shifted into the RSR. One final bit time is measured and the level sampled. This is the

Stop bit, which is always a '1'. If the data recovery circuit samples a '0' in the Stop bit position, then a framing error is set for this character, otherwise the framing error is cleared for this character. See [Receive Framing Error](#) for more information on framing errors.

Immediately after all data bits and the Stop bit have been received, the character in the RSR is transferred to the EUSART receive FIFO, and the EUSART Receive Interrupt Flag (RCxIF) bit of the PIRx register is set. The top character in the FIFO is transferred out of the FIFO by reading the [RCxREG](#) register.



**Important:** If the receive FIFO is overrun, no additional characters will be received until the Overrun condition is cleared. See [Receive Framing Error](#) for more information.

### 31.1.2.3. Receive Interrupts

The EUSART Receive Interrupt Flag (RCxIF) bit of the PIRx register is set whenever the EUSART receiver is enabled and there is an unread character in the receive FIFO. The RCxIF Interrupt Flag bit is read-only, it cannot be set or cleared by software.

RCxIF interrupts are enabled by setting all of the following bits:

- RCxIE, Interrupt Enable bit of the PIRx register
- PEIE, Peripheral Interrupt Enable bit of the INTCON register
- GIE, Global Interrupt Enable bit of the INTCON register

The RCxIF Interrupt Flag bit will be set when there is an unread character in the FIFO, regardless of the state of interrupt enable bits.

### 31.1.2.4. Receive Framing Error

Each character in the receive FIFO buffer has a corresponding framing error Status bit. A framing error indicates that a Stop bit was not seen at the expected time. The framing error status is accessed via the Framing Error ([FERR](#)) bit. The FERR bit represents the status of the top unread character in the receive FIFO. Therefore, the FERR bit must be read before reading the [RCxREG](#) register.

The FERR bit is read-only and only applies to the top unread character in the receive FIFO. A framing error (FERR = 1) does not preclude reception of additional characters. It is not necessary to clear the FERR bit. Reading the next character from the FIFO buffer will advance the FIFO to the next character and the next corresponding framing error.

The FERR bit can be forced clear by clearing the [SPEN](#) bit, which resets the EUSART. Clearing the [CREN](#) bit does not affect the FERR bit. A framing error by itself does not generate an interrupt.



**Important:** If all receive characters in the receive FIFO have framing errors, repeated reads of the RCxREG register will not clear the FERR bit.

### 31.1.2.5. Receive Overrun Error

The receive FIFO buffer can hold two characters. An overrun error will be generated if a third character, in its entirety, is received before the FIFO is accessed. When this happens the Overrun Error ([OERR](#)) bit is set. The characters already in the FIFO buffer can be read but no additional characters will be received until the error is cleared. The error must be cleared by either clearing the [CREN](#) bit or by resetting the EUSART by clearing the [SPEN](#) bit.

### 31.1.2.6. Receiving 9-Bit Characters

The EUSART supports 9-bit character reception. When the 9-Bit Receive Enable ([RX9](#)) bit is set, the EUSART will shift nine bits into the RSR for each character received. The [RX9D](#) bit is the ninth

and Most Significant data bit of the top unread character in the receive FIFO. When reading 9-bit data from the receive FIFO buffer, the RX9D data bit must be read before reading the eight Least Significant bits from the [RCxREG](#) register.

### 31.1.2.7. Address Detection

A special Address Detection mode is available for use when multiple receivers share the same transmission line, such as in RS-485 systems. Address detection is enabled by setting the Address Detect Enable ([ADDEN](#)) bit.

Address detection requires 9-bit character reception. When address detection is enabled, only characters with the ninth data bit set will be transferred to the receive FIFO buffer, thereby setting the RCxIF interrupt bit. All other characters will be ignored.

Upon receiving an address character, user software determines if the address matches its own. Upon address match, user software must disable address detection by clearing the ADDEN bit before the next Stop bit occurs. When user software detects the end of the message, determined by the message protocol used, software places the receiver back into the Address Detection mode by setting the ADDEN bit.

### 31.1.2.8. Asynchronous Reception Setup

1. Initialize the [SPxBRGH:SPxBRGL](#) register pair and the [BRGH](#) and [BRG16](#) bits to achieve the desired baud rate (see [EUSART Baud Rate Generator \(BRG\)](#)).
2. Set the RXxPPS register to select the RXx/DTx input pin.
3. Clear the ANSEL bit for the RXx pin (if applicable).
4. Enable the serial port by setting the [SPEN](#) bit. The [SYNC](#) bit must be cleared for asynchronous operation.
5. If interrupts are desired, set the RCxIE bit of the PEx register and the GIE and PEIE bits of the INTCON register.
6. If 9-bit reception is desired, set the [RX9](#) bit.
7. Enable reception by setting the [CREN](#) bit.
8. The RCxIF Interrupt Flag bit will be set when a character is transferred from the RSR to the receive buffer. An interrupt will be generated if the RCxIE interrupt enable bit was also set.
9. Read the [RCxSTA](#) register to get the Error flags and, if 9-bit data reception is enabled, the ninth data bit.
10. Get the received eight Least Significant data bits from the receive buffer by reading the [RCxREG](#) register.
11. If an overrun occurred, clear the [OERR](#) flag by clearing the CREN receiver enable bit.

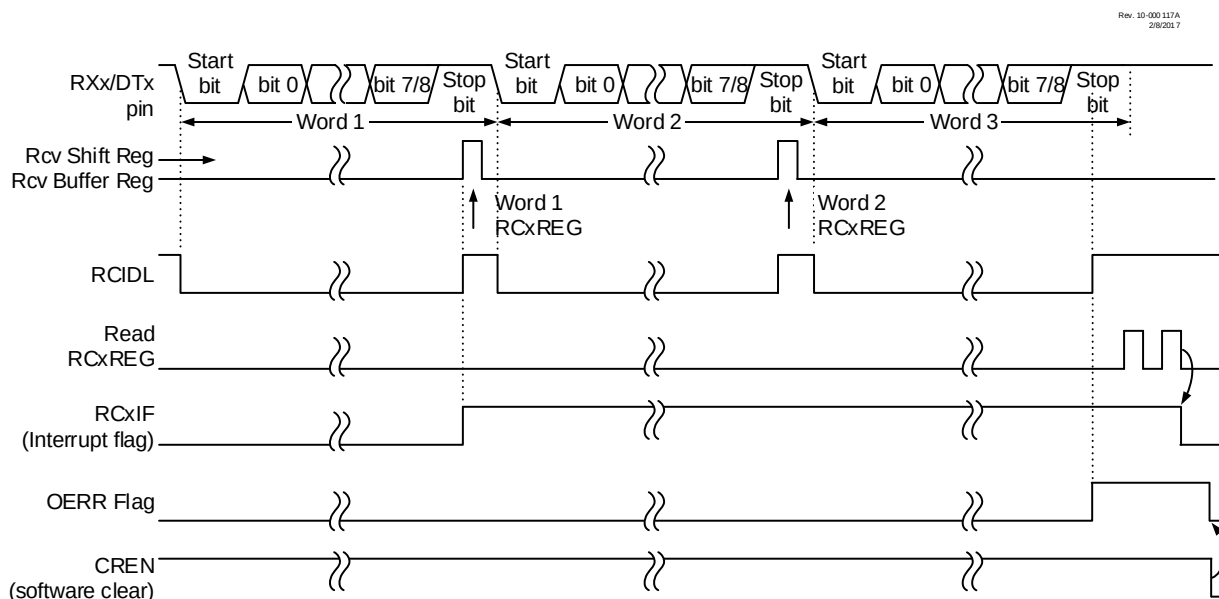
### 31.1.2.9. 9-Bit Address Detection Mode Setup

This mode is typically used in RS-485 systems. To set up an Asynchronous Reception with Address Detect Enable, follow these steps:

1. Initialize the [SPxBRGH:SPxBRGL](#) register pair and the [BRGH](#) and [BRG16](#) bits to achieve the desired baud rate (see [EUSART Baud Rate Generator \(BRG\)](#)).
2. Set the RXxPPS register to select the RXx input pin.
3. Clear the ANSEL bit for the RXx pin (if applicable).
4. Enable the serial port by setting the [SPEN](#) bit. The [SYNC](#) bit must be cleared for asynchronous operation.
5. If interrupts are desired, set the RCxIE bit of the PEx register and the GIE and PEIE bits of the INTCON register.
6. Enable 9-bit reception by setting the [RX9](#) bit.

7. Enable address detection by setting the [ADDEN](#) bit.
8. Enable reception by setting the [CREN](#) bit.
9. The RCxIF Interrupt Flag bit will be set when a character with the ninth bit set is transferred from the RSR to the receive buffer. An interrupt will be generated if the RCxIE interrupt enable bit is also set.
10. Read the [RCxSTA](#) register to get the Error flags. The ninth data bit will always be set.
11. Get the received eight Least Significant data bits from the receive buffer by reading the [RCxREG](#) register. Software determines if this is the device's address.
12. If an overrun occurred, clear the [OERR](#) flag by clearing the CREN receiver enable bit.
13. If the device has been addressed, clear the ADDEN bit to allow all received data into the receive buffer and generate interrupts.

**Figure 31-5. Asynchronous Reception**



**Note:** This timing diagram shows three bytes appearing on the RXx input. The OERR flag is set because the RCxREG register is not read before the third word is received.

## 31.2. Clock Accuracy with Asynchronous Operation

The factory calibrates the internal oscillator block output (INTOSC). However, the INTOSC frequency may drift as  $V_{DD}$  or temperature changes, directly affecting the asynchronous baud rate. Two methods may be used to adjust the baud rate clock, but both require a reference clock source of some kind.

The first (preferred) method uses the OSCTUNE register to adjust the INTOSC output. Adjusting the value in the OSCTUNE register allows for fine resolution changes to the system clock source.

The other method adjusts the value in the Baud Rate Generator. This can be done automatically with the Auto-Baud Detect feature (see [Auto-Baud Detect](#)). There may not be fine enough resolution when adjusting the Baud Rate Generator to compensate for a gradual change in the peripheral clock frequency.

### 31.3. EUSART Baud Rate Generator (BRG)

The Baud Rate Generator (BRG) is an 8-bit or 16-bit timer that is dedicated to the support of both the asynchronous and synchronous EUSART operations. By default, the BRG operates in 8-bit mode. Setting the **BRG16** bit selects 16-bit mode.

The SPxBRGH:SPxBRGL register pair determines the period of the free-running baud rate timer. In Asynchronous mode, the multiplier of the baud rate period is determined by both the **BRGH** and the BRG16 bits. In Synchronous mode, the BRGH bit is ignored.

**Table 31-2** contains the formulas for determining the baud rate. **Equation 31-1** provides a sample calculation for determining the baud rate and baud rate error.

Typical baud rates and error values for various Asynchronous modes have been computed and are shown in the table below. It may be advantageous to use the high baud rate (BRGH = 1) or the 16-bit BRG (BRG16 = 1) to reduce the baud rate error. The 16-bit BRG mode is used to achieve slow baud rates for fast oscillator frequencies. The BRGH bit is used to achieve very high baud rates.

Writing a new value to the SPxBRGH:SPxBRGL register pair causes the BRG timer to be reset (or cleared). This ensures that the BRG does not wait for a timer overflow before outputting the new baud rate.

If the system clock is changed during an active receive operation, a receive error or data loss may result. To avoid this, check the status of the Receive Idle Flag (**RCIDL**) bit to make sure the receive operation is idle before changing the system clock.

#### Equation 31-1. Calculating Baud Rate Error

For a device with  $F_{OSC}$  of 16 MHz, desired baud rate of 9600, Asynchronous mode, 8-bit BRG:

$$DesiredBaudrate = \frac{F_{OSC}}{64 \times (SPxBRG + 1)}$$

Solving for SPxBRG:

$$SPxBRG = \frac{F_{OSC}}{64 \times DesiredBaudrate} - 1$$

$$SPxBRG = \frac{16000000}{64 \times 9600} - 1$$

$$SPxBRG = 25.042 \approx 25$$

$$CalculatedBaudrate = \frac{16000000}{64 \times (25 + 1)}$$

$$CalculatedBaudrate = 9615$$

$$Error = \frac{CalculatedBaudrate - DesiredBaudrate}{DesiredBaudrate}$$

$$Error = \frac{9615 - 9600}{9600}$$

$$Error = 0.16\%$$

**Table 31-2. Baud Rate Formulas**

| Configuration Bits |       |      | BRG/EUSART Mode     | Baud Rate Formula    |
|--------------------|-------|------|---------------------|----------------------|
| SYNC               | BRG16 | BRGH |                     |                      |
| 0                  | 0     | 0    | 8-bit/Asynchronous  | $F_{OSC}/[64 (n+1)]$ |
| 0                  | 0     | 1    | 8-bit/Asynchronous  | $F_{OSC}/[16 (n+1)]$ |
| 0                  | 1     | 0    | 16-bit/Asynchronous |                      |

**Table 31-2. Baud Rate Formulas (continued)**

| Configuration Bits |       |      | BRG/EUSART Mode     | Baud Rate Formula  |
|--------------------|-------|------|---------------------|--------------------|
| SYNC               | BRG16 | BRGH |                     |                    |
| 0                  | 1     | 1    | 16-bit/Asynchronous | $F_{OSC}/[4(n+1)]$ |
| 1                  | 0     | x    | 8-bit/Synchronous   |                    |
| 1                  | 1     | x    | 16-bit/Synchronous  |                    |

**Note:** x = Don't care, n = value of SPxBRGH:SPxBRGL register pair.

**Table 31-3. Sample Baud Rates for Asynchronous Modes**

| Baud Rate | SYNC = 0, BRGH = 0, BRG16 = 0 |         |                       |                               |         |                       |                               |         |                       |                                |         |                       |
|-----------|-------------------------------|---------|-----------------------|-------------------------------|---------|-----------------------|-------------------------------|---------|-----------------------|--------------------------------|---------|-----------------------|
|           | $F_{OSC} = 32.000\text{ MHz}$ |         |                       | $F_{OSC} = 20.000\text{ MHz}$ |         |                       | $F_{OSC} = 18.432\text{ MHz}$ |         |                       | $F_{OSC} = 11.0592\text{ MHz}$ |         |                       |
|           | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                    | % Error | SPBRG value (decimal) |
| 300       | —                             | —       | —                     | —                             | —       | —                     | —                             | —       | —                     | —                              | —       | —                     |
| 1200      | —                             | —       | —                     | 1221                          | 1.73    | 255                   | 1200                          | 0.00    | 239                   | 1200                           | 0.00    | 143                   |
| 2400      | 2404                          | 0.16    | 207                   | 2404                          | 0.16    | 129                   | 2400                          | 0.00    | 119                   | 2400                           | 0.00    | 71                    |
| 9600      | 9615                          | 0.16    | 51                    | 9470                          | -1.36   | 32                    | 9600                          | 0.00    | 29                    | 9600                           | 0.00    | 17                    |
| 10417     | 10417                         | 0.00    | 47                    | 10417                         | 0.00    | 29                    | 10286                         | -1.26   | 27                    | 10165                          | -2.42   | 16                    |
| 19.2k     | 19.23k                        | 0.16    | 25                    | 19.53k                        | 1.73    | 15                    | 19.20k                        | 0.00    | 14                    | 19.20k                         | 0.00    | 8                     |
| 57.6k     | 55.55k                        | -3.55   | 3                     | —                             | —       | —                     | 57.60k                        | 0.00    | 7                     | 57.60k                         | 0.00    | 2                     |
| 115.2k    | —                             | —       | —                     | —                             | —       | —                     | —                             | —       | —                     | —                              | —       | —                     |

| Baud Rate | SYNC = 0, BRGH = 0, BRG16 = 0 |         |                       |                              |         |                       |                               |         |                       |                              |         |                       |
|-----------|-------------------------------|---------|-----------------------|------------------------------|---------|-----------------------|-------------------------------|---------|-----------------------|------------------------------|---------|-----------------------|
|           | $F_{OSC} = 8.000\text{ MHz}$  |         |                       | $F_{OSC} = 4.000\text{ MHz}$ |         |                       | $F_{OSC} = 3.6864\text{ MHz}$ |         |                       | $F_{OSC} = 1.000\text{ MHz}$ |         |                       |
|           | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                  | % Error | SPBRG value (decimal) | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                  | % Error | SPBRG value (decimal) |
| 300       | —                             | —       | —                     | 300                          | 0.16    | 207                   | 300                           | 0.00    | 191                   | 300                          | 0.16    | 51                    |
| 1200      | 1202                          | 0.16    | 103                   | 1202                         | 0.16    | 51                    | 1200                          | 0.00    | 47                    | 1202                         | 0.16    | 12                    |
| 2400      | 2404                          | 0.16    | 51                    | 2404                         | 0.16    | 25                    | 2400                          | 0.00    | 23                    | —                            | —       | —                     |
| 9600      | 9615                          | 0.16    | 12                    | —                            | —       | —                     | 9600                          | 0.00    | 5                     | —                            | —       | —                     |
| 10417     | 10417                         | 0.00    | 11                    | 10417                        | 0.00    | 5                     | —                             | —       | —                     | —                            | —       | —                     |
| 19.2k     | —                             | —       | —                     | —                            | —       | —                     | 19.20k                        | 0.00    | 2                     | —                            | —       | —                     |
| 57.6k     | —                             | —       | —                     | —                            | —       | —                     | 57.60k                        | 0.00    | 0                     | —                            | —       | —                     |
| 115.2k    | —                             | —       | —                     | —                            | —       | —                     | —                             | —       | —                     | —                            | —       | —                     |

| Baud Rate | SYNC = 0, BRGH = 1, BRG16 = 0 |         |                       |                               |         |                       |                               |         |                       |                                |         |                       |
|-----------|-------------------------------|---------|-----------------------|-------------------------------|---------|-----------------------|-------------------------------|---------|-----------------------|--------------------------------|---------|-----------------------|
|           | $F_{OSC} = 32.000\text{ MHz}$ |         |                       | $F_{OSC} = 20.000\text{ MHz}$ |         |                       | $F_{OSC} = 18.432\text{ MHz}$ |         |                       | $F_{OSC} = 11.0592\text{ MHz}$ |         |                       |
|           | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                    | % Error | SPBRG value (decimal) |
| 300       | —                             | —       | —                     | —                             | —       | —                     | —                             | —       | —                     | —                              | —       | —                     |
| 1200      | —                             | —       | —                     | —                             | —       | —                     | —                             | —       | —                     | —                              | —       | —                     |
| 2400      | —                             | —       | —                     | —                             | —       | —                     | —                             | —       | —                     | —                              | —       | —                     |
| 9600      | 9615                          | 0.16    | 207                   | 9615                          | 0.16    | 129                   | 9600                          | 0.00    | 119                   | 9600                           | 0.00    | 71                    |
| 10417     | 10417                         | 0.00    | 191                   | 10417                         | 0.00    | 119                   | 10378                         | -0.37   | 110                   | 10473                          | 0.53    | 65                    |
| 19.2k     | 19.23k                        | 0.16    | 103                   | 19.23k                        | 0.16    | 64                    | 19.20k                        | 0.00    | 59                    | 19.20k                         | 0.00    | 35                    |
| 57.6k     | 57.14k                        | -0.79   | 34                    | 56.82k                        | -1.36   | 21                    | 57.60k                        | 0.00    | 19                    | 57.60k                         | 0.00    | 11                    |

|        |         |      |    |         |       |    |        |      |   |        |      |   |
|--------|---------|------|----|---------|-------|----|--------|------|---|--------|------|---|
| 115.2k | 117.64k | 2.12 | 16 | 113.64k | -1.36 | 10 | 115.2k | 0.00 | 9 | 115.2k | 0.00 | 5 |
|--------|---------|------|----|---------|-------|----|--------|------|---|--------|------|---|

| Baud Rate | SYNC = 0, BRGH = 1, BRG16 = 0 |         |                       |                              |         |                       |                               |         |                       |                              |         |                       |
|-----------|-------------------------------|---------|-----------------------|------------------------------|---------|-----------------------|-------------------------------|---------|-----------------------|------------------------------|---------|-----------------------|
|           | F <sub>OSC</sub> = 8.000 MHz  |         |                       | F <sub>OSC</sub> = 4.000 MHz |         |                       | F <sub>OSC</sub> = 3.6864 MHz |         |                       | F <sub>OSC</sub> = 1.000 MHz |         |                       |
|           | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                  | % Error | SPBRG value (decimal) | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                  | % Error | SPBRG value (decimal) |
| 300       | —                             | —       | —                     | —                            | —       | —                     | —                             | —       | —                     | 300                          | 0.16    | 207                   |
| 1200      | —                             | —       | —                     | 1202                         | 0.16    | 207                   | 1200                          | 0.00    | 191                   | 1202                         | 0.16    | 51                    |
| 2400      | 2404                          | 0.16    | 207                   | 2404                         | 0.16    | 103                   | 2400                          | 0.00    | 95                    | 2404                         | 0.16    | 25                    |
| 9600      | 9615                          | 0.16    | 51                    | 9615                         | 0.16    | 25                    | 9600                          | 0.00    | 23                    | —                            | —       | —                     |
| 10417     | 10417                         | 0.00    | 47                    | 10417                        | 0.00    | 23                    | 10473                         | 0.53    | 21                    | 10417                        | 0.00    | 5                     |
| 19.2k     | 19231                         | 0.16    | 25                    | 19.23k                       | 0.16    | 12                    | 19.2k                         | 0.00    | 11                    | —                            | —       | —                     |
| 57.6k     | 55556                         | -3.55   | 8                     | —                            | —       | —                     | 57.60k                        | 0.00    | 3                     | —                            | —       | —                     |
| 115.2k    | —                             | —       | —                     | —                            | —       | —                     | 115.2k                        | 0.00    | 1                     | —                            | —       | —                     |

| Baud Rate | SYNC = 0, BRGH = 0, BRG16 = 1 |         |                       |                               |         |                       |                               |         |                       |                                |         |                       |
|-----------|-------------------------------|---------|-----------------------|-------------------------------|---------|-----------------------|-------------------------------|---------|-----------------------|--------------------------------|---------|-----------------------|
|           | F <sub>OSC</sub> = 32.000 MHz |         |                       | F <sub>OSC</sub> = 20.000 MHz |         |                       | F <sub>OSC</sub> = 18.432 MHz |         |                       | F <sub>OSC</sub> = 11.0592 MHz |         |                       |
|           | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                    | % Error | SPBRG value (decimal) |
| 300       | 300.0                         | 0.00    | 6666                  | 300.0                         | -0.01   | 4166                  | 300.0                         | 0.00    | 3839                  | 300.0                          | 0.00    | 2303                  |
| 1200      | 1200                          | -0.02   | 3332                  | 1200                          | -0.03   | 1041                  | 1200                          | 0.00    | 959                   | 1200                           | 0.00    | 575                   |
| 2400      | 2401                          | -0.04   | 832                   | 2399                          | -0.03   | 520                   | 2400                          | 0.00    | 479                   | 2400                           | 0.00    | 287                   |
| 9600      | 9615                          | 0.16    | 207                   | 9615                          | 0.16    | 129                   | 9600                          | 0.00    | 119                   | 9600                           | 0.00    | 71                    |
| 10417     | 10417                         | 0.00    | 191                   | 10417                         | 0.00    | 119                   | 10378                         | -0.37   | 110                   | 10473                          | 0.53    | 65                    |
| 19.2k     | 19.23k                        | 0.16    | 103                   | 19.23k                        | 0.16    | 64                    | 19.20k                        | 0.00    | 59                    | 19.20k                         | 0.00    | 35                    |
| 57.6k     | 57.14k                        | -0.79   | 34                    | 56.818                        | -1.36   | 21                    | 57.60k                        | 0.00    | 19                    | 57.60k                         | 0.00    | 11                    |
| 115.2k    | 117.6k                        | 2.12    | 16                    | 113.636                       | -1.36   | 10                    | 115.2k                        | 0.00    | 9                     | 115.2k                         | 0.00    | 5                     |

| Baud Rate | SYNC = 0, BRGH = 0, BRG16 = 1 |         |                       |                              |         |                       |                               |         |                       |                              |         |                       |
|-----------|-------------------------------|---------|-----------------------|------------------------------|---------|-----------------------|-------------------------------|---------|-----------------------|------------------------------|---------|-----------------------|
|           | F <sub>OSC</sub> = 8.000 MHz  |         |                       | F <sub>OSC</sub> = 4.000 MHz |         |                       | F <sub>OSC</sub> = 3.6864 MHz |         |                       | F <sub>OSC</sub> = 1.000 MHz |         |                       |
|           | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                  | % Error | SPBRG value (decimal) | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                  | % Error | SPBRG value (decimal) |
| 300       | 299.9                         | -0.02   | 1666                  | 300.1                        | 0.04    | 832                   | 300.0                         | 0.00    | 767                   | 300.5                        | 0.16    | 207                   |
| 1200      | 1199                          | -0.08   | 416                   | 1202                         | 0.16    | 207                   | 1200                          | 0.00    | 191                   | 1202                         | 0.16    | 51                    |
| 2400      | 2404                          | 0.16    | 207                   | 2404                         | 0.16    | 103                   | 2400                          | 0.00    | 95                    | 2404                         | 0.16    | 25                    |
| 9600      | 9615                          | 0.16    | 51                    | 9615                         | 0.16    | 25                    | 9600                          | 0.00    | 23                    | —                            | —       | —                     |
| 10417     | 10417                         | 0.00    | 47                    | 10417                        | 0.00    | 23                    | 10473                         | 0.53    | 21                    | 10417                        | 0.00    | 5                     |
| 19.2k     | 19.23k                        | 0.16    | 25                    | 19.23k                       | 0.16    | 12                    | 19.20k                        | 0.00    | 11                    | —                            | —       | —                     |
| 57.6k     | 55556                         | -3.55   | 8                     | —                            | —       | —                     | 57.60k                        | 0.00    | 3                     | —                            | —       | —                     |
| 115.2k    | —                             | —       | —                     | —                            | —       | —                     | 115.2k                        | 0.00    | 1                     | —                            | —       | —                     |

| Baud Rate | SYNC = 0, BRGH = 1, BRG16 = 1 or SYNC = 1, BRG16 = 1 |         |                       |                               |         |                       |                               |         |                       |                                |         |                       |
|-----------|------------------------------------------------------|---------|-----------------------|-------------------------------|---------|-----------------------|-------------------------------|---------|-----------------------|--------------------------------|---------|-----------------------|
|           | F <sub>OSC</sub> = 32.000 MHz                        |         |                       | F <sub>OSC</sub> = 20.000 MHz |         |                       | F <sub>OSC</sub> = 18.432 MHz |         |                       | F <sub>OSC</sub> = 11.0592 MHz |         |                       |
|           | Actual Rate                                          | % Error | SPBRG value (decimal) | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                    | % Error | SPBRG value (decimal) |

|        |        |       |       |        |       |       |        |      |       |        |      |      |
|--------|--------|-------|-------|--------|-------|-------|--------|------|-------|--------|------|------|
| 300    | 300.0  | 0.00  | 26666 | 300.0  | 0.00  | 16665 | 300.0  | 0.00 | 15359 | 300.0  | 0.00 | 9215 |
| 1200   | 1200   | 0.00  | 6666  | 1200   | -0.01 | 4166  | 1200   | 0.00 | 3839  | 1200   | 0.00 | 2303 |
| 2400   | 2400   | 0.01  | 3332  | 2400   | 0.02  | 2082  | 2400   | 0.00 | 1919  | 2400   | 0.00 | 1151 |
| 9600   | 9604   | 0.04  | 832   | 9597   | -0.03 | 520   | 9600   | 0.00 | 479   | 9600   | 0.00 | 287  |
| 10417  | 10417  | 0.00  | 767   | 10417  | 0.00  | 479   | 10425  | 0.08 | 441   | 10433  | 0.16 | 264  |
| 19.2k  | 19.18k | -0.08 | 416   | 19.23k | 0.16  | 259   | 19.20k | 0.00 | 239   | 19.20k | 0.00 | 143  |
| 57.6k  | 57.55k | -0.08 | 138   | 57.47k | -0.22 | 86    | 57.60k | 0.00 | 79    | 57.60k | 0.00 | 47   |
| 115.2k | 115.9k | 0.64  | 68    | 116.3k | 0.94  | 42    | 115.2k | 0.00 | 39    | 115.2k | 0.00 | 23   |

| Baud Rate | SYNC = 0, BRGH = 1, BRG16 = 1 or SYNC = 1, BRG16 = 1 |         |                       |                              |         |                       |                               |         |                       |                              |         |                       |
|-----------|------------------------------------------------------|---------|-----------------------|------------------------------|---------|-----------------------|-------------------------------|---------|-----------------------|------------------------------|---------|-----------------------|
|           | F <sub>OSC</sub> = 8.000 MHz                         |         |                       | F <sub>OSC</sub> = 4.000 MHz |         |                       | F <sub>OSC</sub> = 3.6864 MHz |         |                       | F <sub>OSC</sub> = 1.000 MHz |         |                       |
|           | Actual Rate                                          | % Error | SPBRG value (decimal) | Actual Rate                  | % Error | SPBRG value (decimal) | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate                  | % Error | SPBRG value (decimal) |
| 300       | 300.0                                                | 0.00    | 6666                  | 300.0                        | 0.01    | 3332                  | 300.0                         | 0.00    | 3071                  | 300.1                        | 0.04    | 832                   |
| 1200      | 1200                                                 | -0.02   | 1666                  | 1200                         | 0.04    | 832                   | 1200                          | 0.00    | 767                   | 1202                         | 0.16    | 207                   |
| 2400      | 2401                                                 | 0.04    | 832                   | 2398                         | 0.08    | 416                   | 2400                          | 0.00    | 383                   | 2404                         | 0.16    | 103                   |
| 9600      | 9615                                                 | 0.16    | 207                   | 9615                         | 0.16    | 103                   | 9600                          | 0.00    | 95                    | 9615                         | 0.16    | 25                    |
| 10417     | 10417                                                | 0       | 191                   | 10417                        | 0.00    | 95                    | 10473                         | 0.53    | 87                    | 10417                        | 0.00    | 23                    |
| 19.2k     | 19.23k                                               | 0.16    | 103                   | 19.23k                       | 0.16    | 51                    | 19.20k                        | 0.00    | 47                    | 19.23k                       | 0.16    | 12                    |
| 57.6k     | 57.14k                                               | -0.79   | 34                    | 58.82k                       | 2.12    | 16                    | 57.60k                        | 0.00    | 15                    | —                            | —       | —                     |
| 115.2k    | 117.6k                                               | 2.12    | 16                    | 111.1k                       | -3.55   | 8                     | 115.2k                        | 0.00    | 7                     | —                            | —       | —                     |

### 31.3.1. Auto-Baud Detect

The EUSART module supports automatic detection and calibration of the baud rate.

In the Auto-Baud Detect (ABD) mode, the clock to the BRG is reversed. Rather than the BRG clocking the incoming RX signal, the RX signal is timing the BRG. The Baud Rate Generator is used to time the period of a received 55h (ASCII "U") which is the Sync character for the LIN bus. The unique feature of this character is that it has five rising edges, including the Stop bit edge.

Setting the Auto-Baud Detect Enable ([ABDEN](#)) bit starts the auto-baud calibration sequence. While the ABD sequence takes place, the EUSART state machine is held in Idle. On the first rising edge of the receive line, after the Start bit, the [SPxBRG](#) register begins counting up using the BRG counter clock as shown in [Figure 31-6](#). The fifth rising edge will occur on the RXx pin at the end of the eighth bit period. At that time, an accumulated value totaling the proper BRG period is left in the SPxBRGH:SPxBRGL register pair, the ABDEN bit is automatically cleared, and the RCxIF interrupt flag is set. The value in the [RCxREG](#) register needs to be read to clear the RCxIF interrupt. RCxREG content may be discarded. When calibrating for modes that do not use the SPxBRGH register, the user can verify that the SPxBRGL register did not overflow by checking for 00h in the SPxBRGH register.

The BRG auto-baud clock is determined by the [BRG16](#) and [BRGH](#) bits, as shown in [Table 31-4](#). During ABD, both the SPxBRGH and SPxBRGL registers are used as a 16-bit counter, independent of the BRG16 bit setting. While calibrating the baud rate period, the SPxBRGH and SPxBRGL registers are clocked at 1/8<sup>th</sup> the BRG base clock rate. The resulting byte measurement is the average bit time when clocked at full speed.

#### Notes:

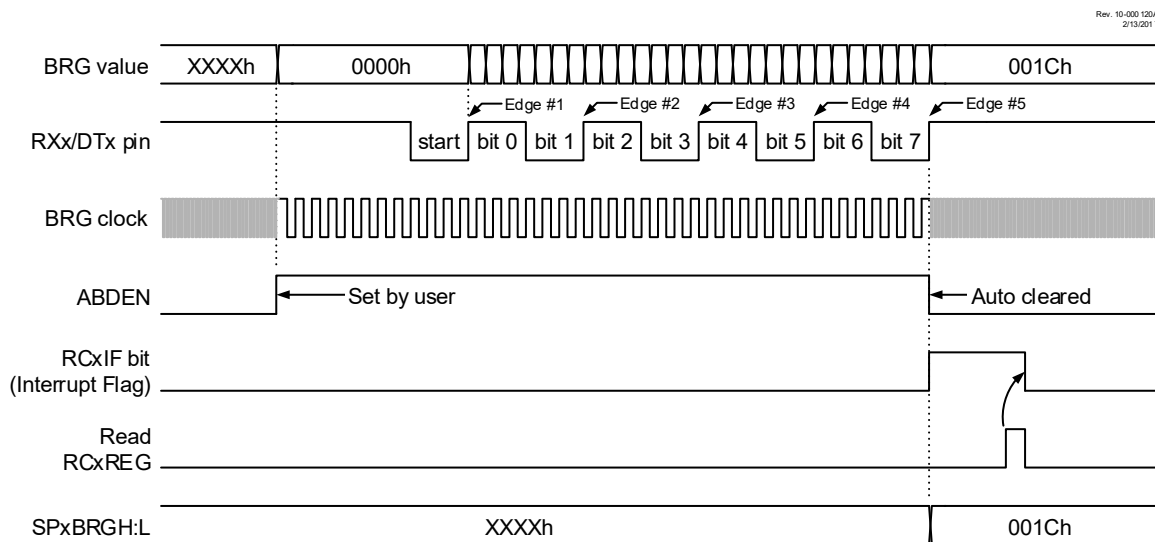
1. If the Wake-Up Enable ([WUE](#)) bit is set with the ABDEN bit, auto-baud detection will occur on the byte following the Break character (see [Auto-Wake-Up on Break](#)).
2. It is up to the user to determine that the incoming character baud rate is within the range of the selected BRG clock source. Some combinations of oscillator frequency and EUSART baud rates are not possible.
3. During the auto-baud process, the auto-baud counter starts counting at one. Upon completion of the auto-baud sequence, to achieve maximum accuracy, subtract 1 from the SPxBRGH:SPxBRGL register pair.

**Table 31-4.** BRG Counter Clock Rates

| BRG16 | BRGH | BRG Base Clock | BRG ABD Clock |
|-------|------|----------------|---------------|
| 1     | 1    | $F_{OSC}/4$    | $F_{OSC}/32$  |
| 1     | 0    | $F_{OSC}/16$   | $F_{OSC}/128$ |
| 0     | 1    | $F_{OSC}/16$   | $F_{OSC}/128$ |
| 0     | 0    | $F_{OSC}/64$   | $F_{OSC}/512$ |

**Note:** During the ABD sequence, the SPxBRGL and SPxBRGH registers are both used as a 16-bit counter, independent of the BRG16 setting.

**Figure 31-6.** Automatic Baud Rate Calibration



### 31.3.2. Auto-Baud Overflow

During the course of automatic baud detection, the Auto-Baud Detect Overflow ([ABDOVF](#)) bit will be set if the baud rate counter overflows before the fifth rising edge is detected on the RXx pin. The ABDOVF bit indicates that the counter has exceeded the maximum count that can fit in the 16 bits of the [SPxBRGH:SPxBRGL](#) register pair. After the ABDOVF bit has been set, the counter continues to count until the fifth rising edge is detected on the RXx pin. Upon detecting the fifth RX edge, the hardware will set the RCxIF interrupt flag and clear the [ABDEN](#) bit. The RCxIF flag can be subsequently cleared by reading the [RCxREG](#) register. The ABDOVF bit can be cleared by software directly.

To terminate the auto-baud process before the RCxIF flag is set, clear the ABDEN bit then clear the ABDOVF bit. The ABDOVF bit will remain set if the ABDEN bit is not cleared first.

### 31.3.3. Auto-Wake-Up on Break

During Sleep mode, all clocks to the EUSART are suspended. Because of this, the Baud Rate Generator is inactive and a proper character reception cannot be performed. The Auto-Wake-Up feature allows the controller to wake up due to activity on the RX/DT line. This feature is available only in Asynchronous mode.

The Auto-Wake-Up feature is enabled by setting the **WUE** bit. Once set, the normal receive sequence on RX/DT is disabled, and the EUSART remains in an Idle state, monitoring for a wake-up event independent of the CPU mode. A wake-up event consists of a high-to-low transition on the RX/DT line. This coincides with the start of a Sync Break or a wake-up signal character for the LIN protocol.

The EUSART module generates an RCxIF interrupt coincident with the wake-up event. The interrupt is generated synchronously to the Q clocks in normal CPU operating modes, as shown in [Figure 31-7](#), and asynchronously if the device is in Sleep mode, as shown in [Figure 31-8](#). The Interrupt condition is cleared by reading the **RCxREG** register.

The WUE bit is automatically cleared by the low-to-high transition on the RX line at the end of the Break. This signals to the user that the Break event is over. At this point, the EUSART module is in Idle mode waiting to receive the next character.

#### 31.3.3.1. Special Considerations

##### Break Character

To avoid character errors or character fragments during a wake-up event, the wake-up character must be all zeros.

When the wake-up is enabled, the function works independent of the low time on the data stream. If the **WUE** bit is set and a valid nonzero character is received, the low time from the Start bit to the first rising edge will be interpreted as the wake-up event. The remaining bits in the character will be received as a fragmented character and subsequent characters can result in framing or overrun errors.

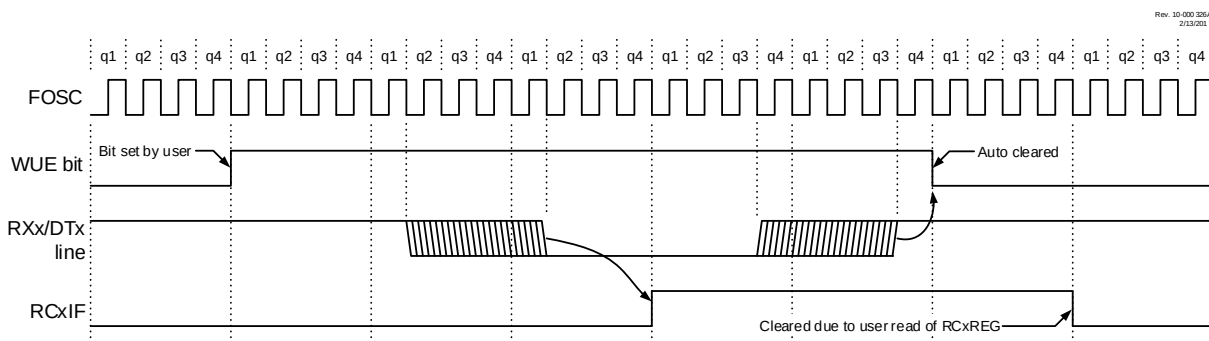
Therefore, the initial character in the transmission must be all '0's. This must be 10 or more bit times, 13-bit times recommended for LIN bus, or any number of bit times for standard RS-232 devices.

##### WUE Bit

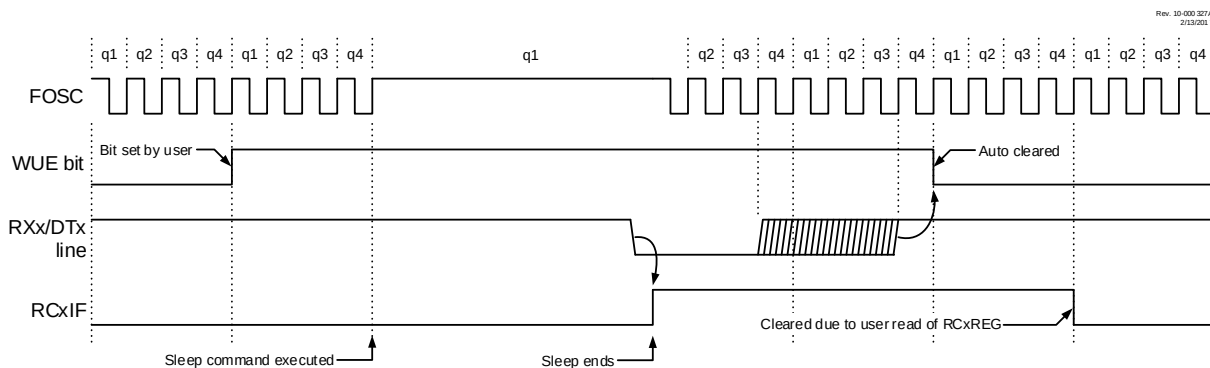
The wake-up event causes a receive interrupt by setting the RCxIF bit. The WUE bit is cleared in hardware by a rising edge on RX/DT. The Interrupt condition is then cleared in software by reading the **RCxREG** register and discarding its contents.

To ensure that no actual data are lost, check the **RCIDL** bit to verify that a receive operation is not in process before setting the WUE bit. If a receive operation is not occurring, the WUE bit may then be set just prior to entering the Sleep mode.

**Figure 31-7.** Auto-Wake-Up (WUE) Bit Timing During Normal Operation



**Note:** The EUSART remains in Idle while the WUE bit is set.

**Figure 31-8.** Auto-Wake-Up (WUE) Bit Timings During Sleep

**Note:** The EUSART remains in Idle while the WUE bit is set.

### 31.3.4. Break Character Sequence

The EUSART module has the capability of sending the special Break character sequences that are required by the LIN bus standard. A Break character consists of a Start bit, followed by 12 '0' bits and a Stop bit.

To send a Break character, set the Send Break Character (**SENDB**) and Transmit Enable (**TXEN**) bits. The Break character transmission is then initiated by a write to the **TXxREG**. The value of data written to **TXxREG** will be ignored and all '0's will be transmitted.

The **SENDB** bit is automatically reset by hardware after the corresponding Stop bit is sent. This allows the user to preload the transmit FIFO with the next transmit byte following the Break character (typically, the Sync character in the LIN specification).

The Transmit Shift Register Status (**TRMT**) bit indicates when the transmit operation is Active or Idle, just as it does during normal transmission. See [Figure 31-9](#) for more details.

#### 31.3.4.1. Break and Sync Transmit Sequence

The following sequence will start a message frame header made up of a Break, followed by an auto-baud Sync byte. This sequence is typical of a LIN bus host.

1. Configure the EUSART for the desired mode.
2. Set the **TXEN** and **SENDB** bits to enable the Break sequence.
3. Load the **TXxREG** with a dummy character to initiate transmission (the value is ignored).
4. Write '55h' to **TXxREG** to load the Sync character into the transmit FIFO buffer.
5. After the Break has been sent, the **SENDB** bit is reset by hardware and the Sync character is then transmitted.

When the **TXxREG** becomes empty, as indicated by **TXxIF**, the next data byte can be written to **TXxREG**.

### 31.3.5. Receiving a Break Character

The EUSART module can receive a Break character in two ways.

The first method to detect a Break character uses the Framing Error (**FERR**) bit and the received data as indicated by **RCxREG**. The Baud Rate Generator is assumed to have been initialized to the expected baud rate.

A Break character has been received when all three of the following conditions are true:

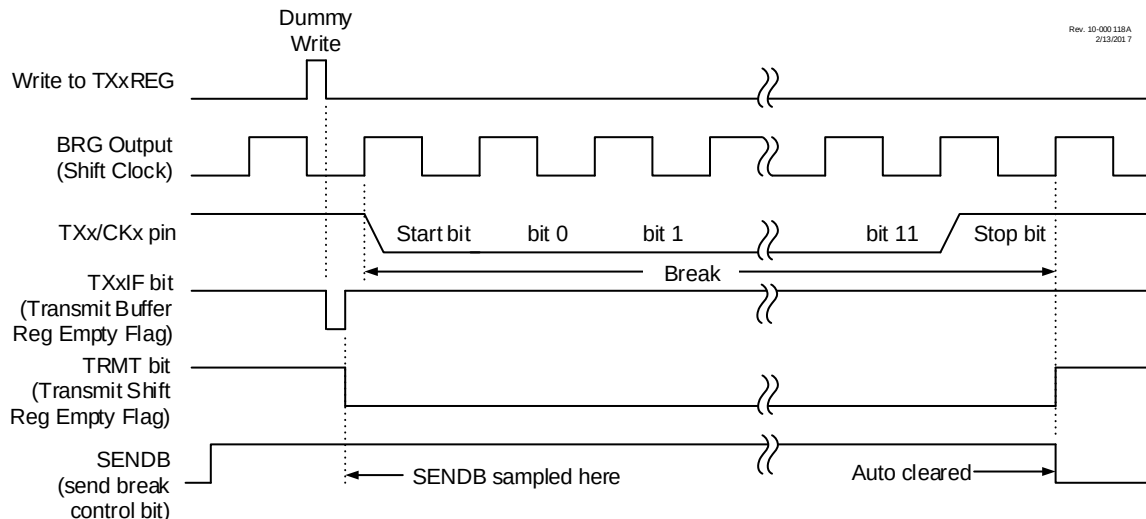
- **RCxIF** bit is set

- FERR bit is set
- RCxREG = 00h

The second method uses the Auto-Wake-Up feature described in [Auto-Wake-Up on Break](#). By enabling this feature, the EUSART will sample the next two transitions on RX/DT, cause an RCxIF interrupt, and receive the next data byte followed by another interrupt.

Note that following a Break character, the user will typically want to enable the Auto-Baud Detect feature. For both methods, the user can set the [ABDEN](#) bit before placing the EUSART in Sleep mode.

**Figure 31-9.** Send Break Character Sequence



## 31.4. EUSART Synchronous Mode

Synchronous serial communications are typically used in systems with a single host and one or more clients. The host device contains the necessary circuitry for baud rate generation and supplies the clock for all devices in the system. Client devices can take advantage of the host clock by eliminating the internal clock generation circuitry.

There are two signal lines in Synchronous mode: A bidirectional data line (DT) and a clock line (CK). The clients use the external clock supplied by the host to shift the serial data into and out of their respective receive and transmit shift registers. Since the data line is bidirectional, synchronous operation is half-duplex only. Half-duplex refers to the fact that host and client devices can receive and transmit data but not both simultaneously. The EUSART can operate as either a host or client device.

Start and Stop bits are not used in synchronous transmissions.

### 31.4.1. Synchronous Host Mode

The following bits are used to configure the EUSART for synchronous host operation:

- The [SYNC](#) bit is set to '1' to configure the EUSART for synchronous operation
- The Clock Source Select ([CSRC](#)) bit is set to '1' to configure the EUSART as the host
- The Single Receive Enable ([SREN](#)) bit is set to '0' for transmit; SREN = 1 for receive (recommended setting to receive 1 byte)
- The Continuous Receive Enable ([CREN](#)) bit is set to '0' for transmit; CREN = 1 to receive continuously

- The **SPEN** bit is set to '1' to enable the EUSART interface



**Important:** Clearing the SREN and CREN bits ensure that the device is in the Transmit mode, otherwise the device will be configured to receive.

#### 31.4.1.1. Host Clock

Synchronous data transfers use a separate clock line, which is synchronous with the data. A device configured as a host transmits the clock on the TX/CK line. The TXx/CKx pin output driver is automatically enabled when the EUSART is configured for synchronous transmit or receive operation. Serial data bits change on the leading edge to ensure they are valid at the trailing edge of each clock. One clock cycle is generated for each data bit. Only as many clock cycles are generated as there are data bits.

#### 31.4.1.2. Clock Polarity

A clock polarity option is provided for Microwire compatibility. Clock polarity is selected with the Clock/Transmit Polarity Select (**SCKP**) bit. Setting the SCKP bit sets the clock Idle state as high. When the SCKP bit is set, the data changes on the falling edge of each clock. Clearing the SCKP bit sets the Idle state as low. When the SCKP bit is cleared, the data changes on the rising edge of each clock.

#### 31.4.1.3. Synchronous Host Transmission

Data are transferred out of the device on the RXx/DTx pin. The RXx/DTx and TXx/CKx pin output drivers are automatically enabled when the EUSART is configured for synchronous host transmit operation.

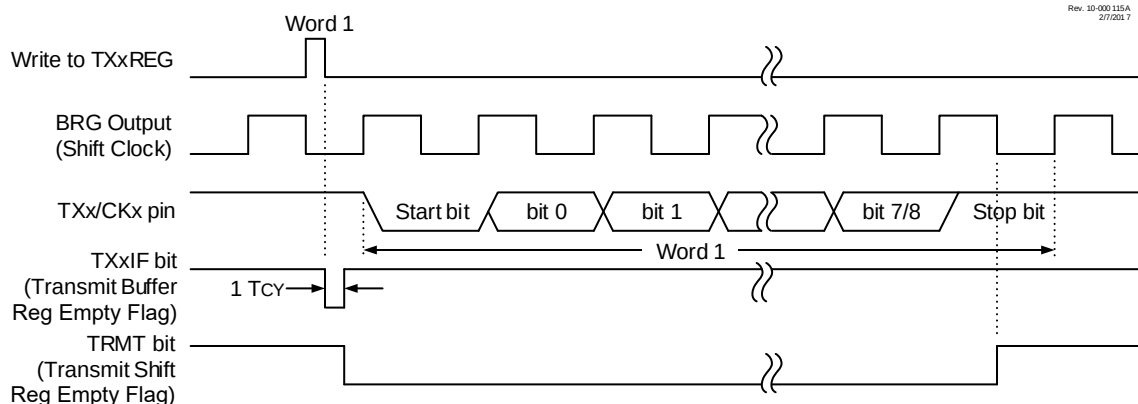
A transmission is initiated by writing a character to the **TXxREG** register. If the TSR still contains all or part of a previous character the new character data are held in the TXxREG until the last bit of the previous character has been transmitted. If this is the first character, or the previous character has been completely flushed from the TSR, the data in the TXxREG is immediately transferred to the TSR. The transmission of the character commences immediately following the transfer of the data to the TSR from the TXxREG.

Each data bit changes on the leading edge of the host clock and remains valid until the subsequent leading clock edge.

**Note:** The TSR register is not mapped in data memory, so it is not available to the user.

#### 31.4.1.4. Synchronous Host Transmission Setup

1. Initialize the **SPxBRGH;SPxBRGL** register pair and the **BRG16** bit to achieve the desired baud rate (see **EUSART Baud Generator (BRG)**).
2. Select the transmit output pin by writing the appropriate values to the RxyPPS register and RXxPPS register. Both selections may enable the same pin.
3. Select the clock output pin by writing the appropriate values to the RxyPPS register and TXxPPS register. Both selections may enable the same pin.
4. Enable the synchronous host serial port by setting bits **SYNC**, **SPEN** and **CSRC**.
5. Disable Receive mode by clearing the **SREN** and **CREN** bits.
6. Enable Transmit mode by setting the **TXEN** bit.
7. If 9-bit transmission is desired, set the **TX9** bit.
8. If interrupts are desired, set the TXxIE bit of the PEx register and the GIE and PEIE bits of the INTCON register.
9. If 9-bit transmission is selected, the ninth bit will be loaded in the **TX9D** bit.
10. Start transmission by loading data to the **TXxREG** register.

**Figure 31-10. Synchronous Transmission**

### 31.4.1.5. Synchronous Host Reception

Data are received at the RXx/DTx pin. The RXx/DTx pin output driver is automatically disabled when the EUSART is configured for synchronous host receive operation.

In Synchronous mode, reception is enabled by setting either the Single Receive Enable (**SREN**) bit or the Continuous Receive Enable (**CREN**) bit.

When SREN is set and CREN is clear, only as many clock cycles are generated as there are data bits in a single character. The SREN bit is automatically cleared at the completion of one character. When CREN is set, clocks are continuously generated until CREN is cleared. If CREN is cleared in the middle of a character the CK clock stops immediately and the partial character is discarded. If SREN and CREN are both set, then SREN is cleared at the completion of the first character and CREN takes precedence.

To initiate reception, set either SREN or CREN. Data are sampled at the RXx/DTx pin on the trailing edge of the TX/CK clock pin and is shifted into the Receive Shift Register (RSR). When a complete character is received into the RSR, the RCxIF bit is set and the character is automatically transferred to the two character receive FIFO. The eight Least Significant bits of the top character in the receive FIFO are available in **RCxREG**. The RCxIF bit remains set as long as there are unread characters in the receive FIFO.

**Note:** If the RX/DT function is on an analog pin, the corresponding ANSEL bit must be cleared for the receiver to function.

### 31.4.1.6. Client Clock

Synchronous data transfers use a separate clock line, which is synchronous with the data. A device configured as a client receives the clock on the TX/CK line. The TXx/CKx pin output driver is automatically disabled when the device is configured for synchronous client transmit or receive operation. Serial data bits change on the leading edge to ensure they are valid at the trailing edge of each clock. One data bit is transferred for each clock cycle. Only as many clock cycles may be received as there are data bits.



**Important:** If the device is configured as a client and the TX/CK function is on an analog pin, the corresponding ANSEL bit must be cleared.

### 31.4.1.7. Receive Overrun Error

The receive FIFO buffer can hold two characters. An overrun error will be generated if a third character, in its entirety, is received before the FIFO is accessed. When this happens the Overrun Error (**OERR**) bit is set. The characters already in the FIFO buffer can be read but no additional

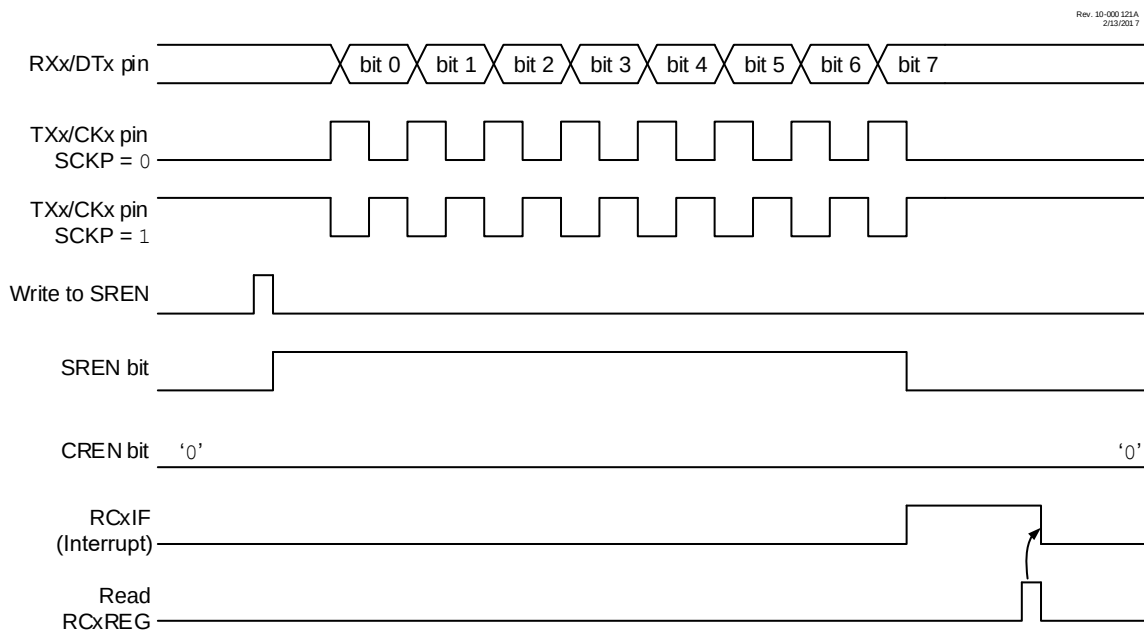
characters will be received until the error is cleared. The error must be cleared by either clearing the **CREN** bit or by resetting the EUSART by clearing the **SPEN** bit.

#### 31.4.1.8. Receiving 9-Bit Characters

The EUSART supports 9-bit character reception. When the 9-Bit Receive Enable (**RX9**) bit is set, the EUSART will shift nine bits into the RSR for each character received. The **RX9D** bit is the ninth and Most Significant data bit of the top unread character in the receive FIFO. When reading 9-bit data from the receive FIFO buffer, the **RX9D** data bit must be read before reading the eight Least Significant bits from the **RCxREG** register.

#### 31.4.1.9. Synchronous Host Reception Setup

1. Initialize the **SPxBRGH:SPxBRGL** register pair and set or clear the **BRG16** bit, as required, to achieve the desired baud rate.
2. Select the receive input pin by writing the appropriate values to the **RxyPPS** and **RXxPPS** registers. Both selections may enable the same pin.
3. Select the clock output pin by writing the appropriate values to the **RxyPPS** and **TXxPPS** registers. Both selections may enable the same pin.
4. Clear the **ANSEL** bit for the **RXx** pin (if applicable).
5. Enable the synchronous host serial port by setting the **SYNC**, **SPEN** and **CSRC** bits.
6. Ensure that the **CREN** and **SREN** bits are cleared.
7. If interrupts are desired, set the **RCxIE** bit of the **PIEx** register and the **GIE** and **PEIE** bits of the **INTCON** register.
8. If 9-bit reception is desired, set the **RX9** bit.
9. Start reception by setting the **SREN** bit or, for continuous reception, set the **CREN** bit.
10. The **RCxIF** Interrupt Flag bit will be set when reception of a character is complete. An interrupt will be generated if the **RCxIE** enable bit was set.
11. Read the **RCxSTA** register to get the ninth bit (if enabled) and determine if any error occurred during reception.
12. Read the 8-bit received data by reading the **RCxREG** register.
13. If an overrun error occurs, clear the error by either clearing the **CREN** bit or by clearing the **SPEN** bit which resets the EUSART.

**Figure 31-11.** Synchronous Reception (Host Mode, SREN)

### 31.4.2. Synchronous Client Mode

The following bits are used to configure the EUSART for synchronous client operation:

- **SYNC** = 1 (configures the EUSART for synchronous operation)
- **CSRC** = 0 (configures the EUSART as a client)
- **SREN** = 0 (for transmit); **SREN** = 1 (for single byte receive)
- **CREN** = 0 (for transmit); **CREN** = 1 (recommended setting for continuous receive)
- **SPEN** = 1 (enables the EUSART)



**Important:** Clearing the SREN and CREN bits ensure that the device is in Transmit mode, otherwise the device will be configured to receive.

#### 31.4.2.1. EUSART Synchronous Client Transmit

The operation of the Synchronous Host and Client modes are identical (see [Synchronous Host Transmission](#)), except in the case of the Sleep mode.

If two words are written to the **TXxREG** and then the **SLEEP** instruction is executed, the following will occur:

1. The first character will immediately transfer to the TSR register and transmit.
2. The second word will remain in the TXxREG register.
3. The TXxIF bit will not be set.
4. After the first character has been shifted out of TSR, the TXxREG register will transfer the second character to the TSR and the TXxIF bit will now be set.
5. If the PEIE and TXxIE bits are set, the interrupt will wake the device from Sleep and execute the next instruction. If the GIE bit is also set, the program will call the Interrupt Service Routine.

### 31.4.2.2. Synchronous Client Transmission Setup

1. Set the [SYNC](#) and [SPEN](#) bits and clear the [CSRC](#) bit.
2. Select the transmit output pin by writing the appropriate values to the RxyPPS register and RXxPPS register. Both selections may enable the same pin.
3. Select the clock input pin by writing the appropriate value to the TXxPPS register.
4. Clear the ANSEL bit for the CKx pin (if applicable).
5. Clear the [CREN](#) and [SREN](#) bits.
6. If interrupts are desired, set the TXxIE bit of the PEx register and the GIE and PEIE bits of the INTCON register.
7. If 9-bit transmission is desired, set the [TX9](#) bit.
8. Enable transmission by setting the [TXEN](#) bit.
9. If 9-bit transmission is selected, insert the Most Significant bit into the [TX9D](#) bit.
10. Prepare for transmission by writing the eight Least Significant bits to the [TXxREG](#) register. The word will be transmitted in response to the Host clocks at the CKx pin.

### 31.4.2.3. EUSART Synchronous Client Reception

The operation of the Synchronous Host and Client modes is identical (see [Synchronous Host Reception](#)), with the following exceptions:

- Sleep
- [CREN](#) bit is always set, therefore the receiver is never Idle
- [SREN](#) bit, which is a “don’t care” in Client mode

A character may be received while in Sleep mode by setting the CREN bit prior to entering Sleep. Once the word is received, the RSR register will transfer the data to the [RCxREG](#) register. If the RCxIE enable bit is set, the interrupt generated will wake the device from Sleep and execute the next instruction. If the GIE bit is also set, the program will branch to the interrupt vector.

### 31.4.2.4. Synchronous Client Reception Setup

1. Set the [SYNC](#) and [SPEN](#) bits and clear the [CSRC](#) bit.
2. Select the receive input pin by writing the appropriate value to the RXxPPS register.
3. Select the clock input pin by writing the appropriate values to the TXxPPS register.
4. Clear the ANSEL bit for both the TXx/CKx and RXx/DTx pins (if applicable).
5. If interrupts are desired, set the RCxIE bit of the PEx register and the GIE and PEIE bits of the INTCON register.
6. If 9-bit reception is desired, set the [RX9](#) bit.
7. Set the [CREN](#) bit to enable reception.
8. The RCxIF bit will be set when reception is complete. An interrupt will be generated if the RCxIE bit was set.
9. If 9-bit mode is enabled, retrieve the Most Significant bit from the [RX9D](#) bit.
10. Retrieve the eight Least Significant bits from the receive FIFO by reading the [RCxREG](#) register.
11. If an overrun error occurs, clear the error by either clearing the CREN bit or by clearing the SPEN bit which resets the EUSART.

## 31.5. EUSART Operation During Sleep

The EUSART will remain active during Sleep only in the Synchronous Client mode. All other modes require the system clock and therefore cannot generate the necessary signals to run the Transmit or Receive Shift registers during Sleep.

Synchronous Client mode uses an externally generated clock to run the Transmit and Receive Shift registers.

### 31.5.1. Synchronous Receive During Sleep

To receive during Sleep, all the following conditions must be met before entering Sleep mode:

- [RCxSTA](#) and [TxSTA](#) Control registers must be configured for Synchronous Client Reception (see [Synchronous Client Reception Setup](#)).
- If interrupts are desired, set the RCxIE bit of the PIRx register and the GIE and PEIE bits of the INTCON register.
- The RCxIF interrupt flag must be cleared by reading [RCxREG](#) to unload any pending characters in the receive buffer.

Upon entering Sleep mode, the device will be ready to accept data and clocks on the RXx/DTx and TXx/CKx pins, respectively. When the data word has been completely clocked in by the external device, the RCxIF Interrupt Flag bit of the PIRx register will be set. Thereby, waking the processor from Sleep.

Upon waking from Sleep, the instruction following the `SLEEP` instruction will be executed. If the Global Interrupt Enable (GIE) bit of the INTCON register is also set, then the Interrupt Service Routine (ISR) will be called.

### 31.5.2. Synchronous Transmit During Sleep

To transmit during Sleep, the following conditions must be met before entering Sleep mode:

- The [RCxSTA](#) and [TxSTA](#) Control registers must be configured for synchronous client transmission (see [Synchronous Client Transmission Setup](#)).
- The TXxIF interrupt flag must be cleared by writing the output data to the [TxREG](#), thereby filling the TSR and transmit buffer.
- The TXxIE interrupt enable bits of the PIRx register and PEIE of the INTCON register must be written to '1'.
- If interrupts are desired, set the GIE bit of the INTCON register.

Upon entering Sleep mode, the device will be ready to accept clocks on the TXx/CKx pin and transmit data on the RXx/DTx pin. When the data word in the TSR register has been completely clocked out by the external device, the pending byte in the TXxREG will transfer to the TSR and the TXxIF flag will be set. Thereby, waking the processor from Sleep. At this point, the TXxREG is available to accept another character for transmission. Writing TXxREG will clear the TXxIF flag.

Upon waking from Sleep, the instruction following the `SLEEP` instruction will be executed. If the Global Interrupt Enable (GIE) bit is also set then the Interrupt Service Routine (ISR) will be called.

## 31.6. Register Definitions: EUSART Control

**31.6.1. TXxSTA****Name:** TXxSTA**Offset:** 0x0711

Transmit Status and Control Register

| Bit    | 7    | 6   | 5    | 4    | 3     | 2    | 1    | 0    |
|--------|------|-----|------|------|-------|------|------|------|
|        | CSRC | TX9 | TXEN | SYNC | SENDB | BRGH | TRMT | TX9D |
| Access | R/W  | R/W | R/W  | R/W  | R/W   | R/W  | R    | R/W  |
| Reset  | 0    | 0   | 0    | 0    | 0     | 0    | 1    | 0    |

**Bit 7 – CSRC** Clock Source Select

| Value | Condition | Description                                     |
|-------|-----------|-------------------------------------------------|
| 1     | SYNC = 1  | Host mode (clock generated internally from BRG) |
| 0     | SYNC = 1  | Client mode (clock from external source)        |
| x     | SYNC = 0  | Don't care                                      |

**Bit 6 – TX9** 9-Bit Transmit Enable

| Value | Description                |
|-------|----------------------------|
| 1     | Selects 9-bit transmission |
| 0     | Selects 8-bit transmission |

**Bit 5 – TXEN** Transmit EnableEnables transmitter<sup>(1)</sup>

| Value | Description       |
|-------|-------------------|
| 1     | Transmit enabled  |
| 0     | Transmit disabled |

**Bit 4 – SYNC** EUSART Mode Select

| Value | Description       |
|-------|-------------------|
| 1     | Synchronous mode  |
| 0     | Asynchronous mode |

**Bit 3 – SENDB** Send Break Character

| Value | Condition | Description                                                                |
|-------|-----------|----------------------------------------------------------------------------|
| 1     | SYNC = 0  | Send Sync Break on next transmission (cleared by hardware upon completion) |
| 0     | SYNC = 0  | Sync Break transmission disabled or completed                              |
| x     | SYNC = 1  | Don't care                                                                 |

**Bit 2 – BRGH** High Baud Rate Select

| Value | Condition | Description                                                       |
|-------|-----------|-------------------------------------------------------------------|
| 1     | SYNC = 0  | High speed, if BRG16 = 1, baud rate is baudclk/4; else baudclk/16 |
| 0     | SYNC = 0  | Low speed                                                         |
| x     | SYNC = 1  | Don't care                                                        |

**Bit 1 – TRMT** Transmit Shift Register (TSR) Status

| Value | Description      |
|-------|------------------|
| 1     | TSR is empty     |
| 0     | TSR is not empty |

**Bit 0 – TX9D** Ninth Bit of Transmit Data

Can be address/data bit or a parity bit.

**Note:** 1. The [SREN](#) and [CREN](#) bits override TXEN in Sync mode.

**31.6.2. RCxSTA****Name:** RCxSTA**Offset:** 0x0710

Receive Status and Control Register

| Bit    | 7    | 6   | 5      | 4    | 3     | 2    | 1    | 0    |
|--------|------|-----|--------|------|-------|------|------|------|
|        | SPEN | RX9 | SREN   | CREN | ADDEN | FERR | OERR | RX9D |
| Access | R/W  | R/W | R/W/HC | R/W  | R/W   | R    | R/HC | R/HC |
| Reset  | 0    | 0   | 0      | 0    | 0     | 0    | 0    | 0    |

**Bit 7 – SPEN** Serial Port Enable

| Value | Description                          |
|-------|--------------------------------------|
| 1     | Serial port enabled                  |
| 0     | Serial port disabled (held in Reset) |

**Bit 6 – RX9** 9-Bit Receive Enable

| Value | Description             |
|-------|-------------------------|
| 1     | Selects 9-bit reception |
| 0     | Selects 8-bit reception |

**Bit 5 – SREN** Single Receive Enable

Controls reception. This bit is cleared by hardware when reception is complete

| Value | Condition                                      | Description                |
|-------|------------------------------------------------|----------------------------|
| 1     | $\text{SYNC} = 1 \text{ AND } \text{CSRC} = 1$ | Start single receive       |
| 0     | $\text{SYNC} = 1 \text{ AND } \text{CSRC} = 1$ | Single receive is complete |
| x     | $\text{SYNC} = 0 \text{ OR } \text{CSRC} = 0$  | Don't care                 |

**Bit 4 – CREN** Continuous Receive Enable

| Value | Condition         | Description                                                                       |
|-------|-------------------|-----------------------------------------------------------------------------------|
| 1     | $\text{SYNC} = 1$ | Enables continuous receive until enable bit CREN is cleared (CREN overrides SREN) |
| 0     | $\text{SYNC} = 1$ | Disables continuous receive                                                       |
| 1     | $\text{SYNC} = 0$ | Enables receiver                                                                  |
| 0     | $\text{SYNC} = 0$ | Disables receiver                                                                 |

**Bit 3 – ADDEN** Address Detect Enable

| Value | Condition                                     | Description                                                                                   |
|-------|-----------------------------------------------|-----------------------------------------------------------------------------------------------|
| 1     | $\text{SYNC} = 0 \text{ AND } \text{RX9} = 1$ | The receive buffer is loaded and the interrupt occurs only when the ninth received bit is set |
| 0     | $\text{SYNC} = 0 \text{ AND } \text{RX9} = 1$ | All bytes are received and interrupt always occurs. Ninth bit can be used as parity bit       |
| x     | $\text{RX9} = 0 \text{ OR } \text{SYNC} = 1$  | Don't care                                                                                    |

**Bit 2 – FERR** Framing Error

| Value | Description                                                         |
|-------|---------------------------------------------------------------------|
| 1     | Unread byte in <a href="#">RCxREG</a> has a framing error           |
| 0     | Unread byte in <a href="#">RCxREG</a> does not have a framing error |

**Bit 1 – OERR** Overrun Error

| Value | Description                                                        |
|-------|--------------------------------------------------------------------|
| 1     | Overrun error (can be cleared by clearing either SPEN or CREN bit) |
| 0     | No overrun error                                                   |

**Bit 0 – RX9D** Ninth bit of Received Data

This can be address/data bit or a parity bit which is determined by user firmware.

**31.6.3. BAUDxCON****Name:** BAUDxCON**Offset:** 0x0712

Baud Rate Control Register

| Bit    | 7      | 6     | 5 | 4    | 3     | 2 | 1   | 0     |
|--------|--------|-------|---|------|-------|---|-----|-------|
|        | ABDOVF | RCIDL |   | SCKP | BRG16 |   | WUE | ABDEN |
| Access | R      | R     |   | R/W  | R/W   |   | R/W | R/W   |
| Reset  | 0      | 1     |   | 0    | 0     |   | 0   | 0     |

**Bit 7 – ABDOVF** Auto-Baud Detect Overflow

| Value | Condition | Description                      |
|-------|-----------|----------------------------------|
| 1     | SYNC = 0  | Auto-baud timer overflowed       |
| 0     | SYNC = 0  | Auto-baud timer did not overflow |
| x     | SYNC = 1  | Don't care                       |

**Bit 6 – RCIDL** Receive Idle Flag

| Value | Condition | Description                                               |
|-------|-----------|-----------------------------------------------------------|
| 1     | SYNC = 0  | Receiver is Idle                                          |
| 0     | SYNC = 0  | Start bit has been received and the receiver is receiving |
| x     | SYNC = 1  | Don't care                                                |

**Bit 4 – SCKP** Clock/Transmit Polarity Select

| Value | Condition | Description                                                                   |
|-------|-----------|-------------------------------------------------------------------------------|
| 1     | SYNC = 0  | Idle state for transmit (TX) is a low level (transmit data are inverted)      |
| 0     | SYNC = 0  | Idle state for transmit (TX) is a high level (transmit data are non-inverted) |
| 1     | SYNC = 1  | Data are clocked on rising edge of the clock                                  |
| 0     | SYNC = 1  | Data are clocked on falling edge of the clock                                 |

**Bit 3 – BRG16** 16-bit Baud Rate Generator Select

| Value | Description                        |
|-------|------------------------------------|
| 1     | 16-bit Baud Rate Generator is used |
| 0     | 8-bit Baud Rate Generator is used  |

**Bit 1 – WUE** Wake-Up Enable

| Value | Condition | Description                                                                                                                                                               |
|-------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | SYNC = 0  | Receiver is waiting for a falling edge. Upon falling edge, no character will be received and the RCxIF flag will be set. WUE will automatically clear after RCxIF is set. |
| 0     | SYNC = 0  | Receiver is operating normally                                                                                                                                            |
| x     | SYNC = 1  | Don't care                                                                                                                                                                |

**Bit 0 – ABDEN** Auto-Baud Detect Enable

| Value | Condition | Description                                                          |
|-------|-----------|----------------------------------------------------------------------|
| 1     | SYNC = 0  | Auto-Baud Detect mode is enabled (clears when auto-baud is complete) |
| 0     | SYNC = 0  | Auto-Baud Detect is complete or mode is disabled                     |
| x     | SYNC = 1  | Don't care                                                           |

### 31.6.4. RCxREG

**Name:** RCxREG

**Offset:** 0x070C

Receive Data Register

| Bit    | 7          | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|--------|------------|---|---|---|---|---|---|---|
|        | RCREG[7:0] |   |   |   |   |   |   |   |
| Access | R          | R | R | R | R | R | R | R |
| Reset  | 0          | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**Bits 7:0 – RCREG[7:0]** Receive data

### 31.6.5. TXxREG

**Name:** TXxREG

**Offset:** 0x070D

Transmit Data Register

| Bit    | 7          | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
|--------|------------|-----|-----|-----|-----|-----|-----|-----|
|        | TXREG[7:0] |     |     |     |     |     |     |     |
| Access | R/W        | R/W | R/W | R/W | R/W | R/W | R/W | R/W |
| Reset  | 0          | 0   | 0   | 0   | 0   | 0   | 0   | 0   |

**Bits 7:0 – TXREG[7:0]** Transmit Data

**31.6.6. SPxBRG****Name:** SPxBRG**Offset:** 0x070E

EUSART Baud Rate Generator

|        |             |     |     |     |     |     |     |     |
|--------|-------------|-----|-----|-----|-----|-----|-----|-----|
| Bit    | 15          | 14  | 13  | 12  | 11  | 10  | 9   | 8   |
|        | SPBRG[15:8] |     |     |     |     |     |     |     |
| Access | R/W         | R/W | R/W | R/W | R/W | R/W | R/W | R/W |
| Reset  | 0           | 0   | 0   | 0   | 0   | 0   | 0   | 0   |
| Bit    | 7           | 6   | 5   | 4   | 3   | 2   | 1   | 0   |
|        | SPBRG[7:0]  |     |     |     |     |     |     |     |
| Access | R/W         | R/W | R/W | R/W | R/W | R/W | R/W | R/W |
| Reset  | 0           | 0   | 0   | 0   | 0   | 0   | 0   | 0   |

**Bits 15:0 – SPBRG[15:0] Baud Rate Register**

**Notes:** The individual bytes in this multibyte register can be accessed with the following register names:

- SPxBRGH: Accesses the high byte SPBRG[15:8]
- SPxBRGL: Accesses the low byte SPBRG[7:0]

### 31.7. Register Summary - EUSART

| Offset                | Name     | Bit Pos. | 7           | 6     | 5    | 4    | 3     | 2    | 1    | 0     |
|-----------------------|----------|----------|-------------|-------|------|------|-------|------|------|-------|
| 0x00<br>...<br>0x070B | Reserved |          |             |       |      |      |       |      |      |       |
| 0x070C                | RC1REG   | 7:0      | RCREG[7:0]  |       |      |      |       |      |      |       |
| 0x070D                | TX1REG   | 7:0      | TXREG[7:0]  |       |      |      |       |      |      |       |
| 0x070E                | SP1BRG   | 7:0      | SPBRG[7:0]  |       |      |      |       |      |      |       |
|                       |          | 15:8     | SPBRG[15:8] |       |      |      |       |      |      |       |
| 0x0710                | RC1STA   | 7:0      | SPEN        | RX9   | SREN | CREN | ADDEN | FERR | OERR | RX9D  |
| 0x0711                | TX1STA   | 7:0      | CSRC        | TX9   | TXEN | SYNC | SENDB | BRGH | TRMT | TX9D  |
| 0x0712                | BAUD1CON | 7:0      | ABDOVF      | RCIDL |      | SCKP | BRG16 |      | WUE  | ABDEN |